Aeromagnetic Survey of the Troodos Igneous Massif, Cyprus

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Aeromagnetic data, combined with other observations, support the view that some of the crust beneath the eastern Mediterranean might be oceanic, formed by seafloor spreading at a time when the Earth's magnetic field was of constant and normal polarity.

PETROLOGICAL, structural, radiometric and geophysical data suggest that the Troodos Massif of Cyprus is an uplifted fragment of deep ocean lithosphere formed by the process of seafloor spreading during the late Mesozoic[1–6]. Indeed, the rock types and structures of the Massif have been quoted as those which fit most closely the model for the deep ocean floor deduced from geological and geophysical studies at sea[7,8]. If this is so, it is of considerable interest to ascertain whether any part of the Massif is characterized by oceanic type, linear magnetic anomalies[9]; for, if present, such anomalies would greatly strengthen the suggestion that the Troodos Massif represents deep seafloor formed by seafloor spreading, and facilitate, for the first time, a detailed and direct test of the Vine-Matthews hypothesis[10] (that the linear anomalies correlate with avenues of normally and reversely magnetized material within the basaltic layer of oceanic crust).

In November 1969 a high level aeromagnetic survey of southern Cyprus and the adjoining offshore areas was made by Hunting Geology and Geophysics Ltd, London. The survey was flown at a constant height of 2.6 km above mean sea level and involved 5,000 km of continuous profiling. South of latitude 35° 15′ N the survey area was traversed by east–west flight lines spaced 4 to 5 km apart, with additional coverage over the Troodos Mountains where the flight line spacing was reduced to 1.7 km. Ten north–south "tie lines" were also flown, spaced at approximately 17 km intervals, and traversing the whole island. Flight path recovery was by means of aerial photography over land and Doppler navigation at sea; the aircraft also carried an hypsometer recorder, to provide a continuous check on altitude, and a caesium vapour magnetometer to measure the Earth's total magnetic field. A fluxgate magnetometer was installed on the ground as a magnetic storm detector.

Magnetic and Bouguer Anomalies

In order to obtain the magnetic anomaly map (Fig. 1) a planar regional gradient, derived from the International Geomagnetic Reference Field[11], was removed from the airborne magnetometer records. This regional field, which takes into account the time and altitude of the survey, is given by the expression $44{,}386+6.2\ N+2\ E$ gamma, where N and E are minutes of latitude and longitude north and east of 34° 30′ N

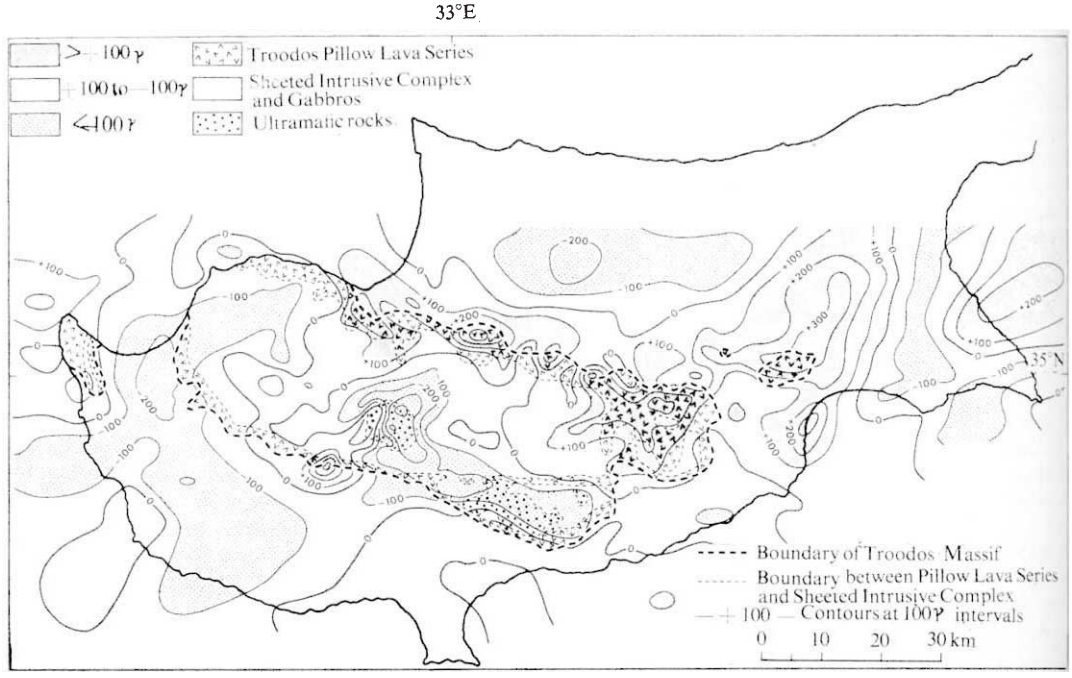


Fig. 1 Resulting magnetic anomaly map of southern Cyprus after removal of an appropriate regional field from the total field aerosurvey (see text).

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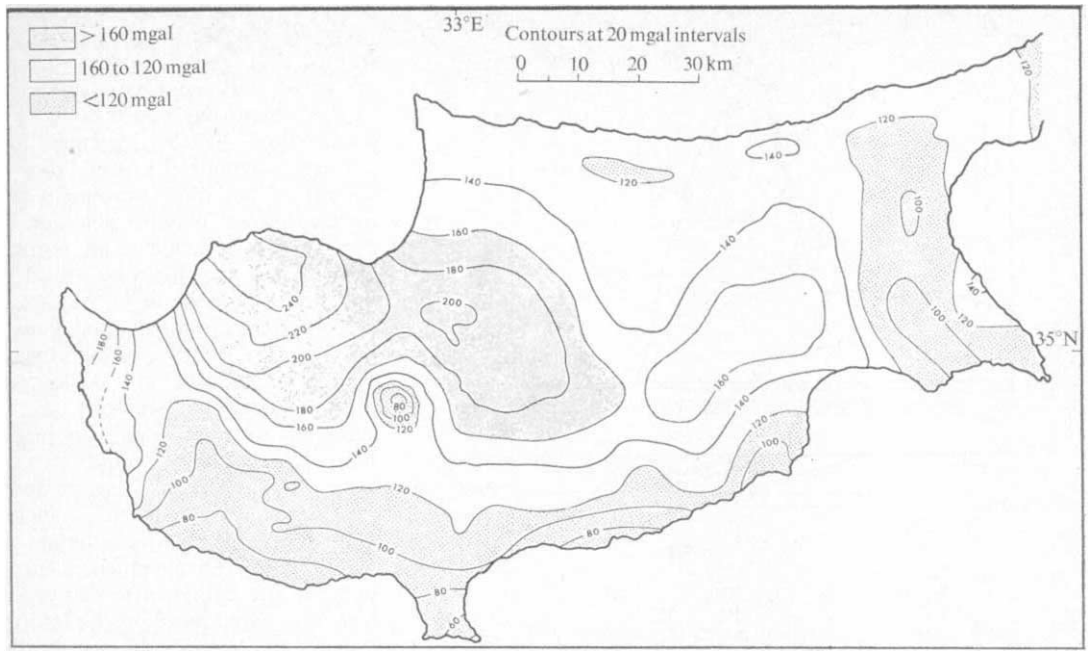


Fig. 2 Bouguer gravity anomaly map of Cyprus, reproduced from Gass and Masson-Smith[1]. 1 mgal = 10^{-5} m s^{-2}.

and 32° 10′ E respectively. No specific correction was made for diurnal variations although these are accounted for, in part, by the procedure for distributing cross-over errors. In the first instance the flight lines and magnetic data were plotted on 1 : 50,000 scale base maps and the magnetic anomalies contoured at intervals of 20 gamma. Subsequently these maps were compiled and reduced to 1 : 100,000 and 1 : 250,000. But, for clarity, only the 100 gamma contours of the anomaly map have been reproduced in Fig. 1. Although the flight-line spacing was a little wider than ideal for the terrain clearance, there is every indication from the resulting anomaly map that the survey gives a very detailed and accurate representation of disturbances in the Earth's magnetic field at this altitude over Cyprus.

As expected, the largest magnetic anomalies and steepest anomaly gradients correlate with the igneous outcrop area of the Troodos Massif, and, in particular, with the outcrop of the peripheral pillow lavas (Fig. 1). Away from the outcropping areas of the Massif, there is a remarkably close correlation between the magnetic anomaly map and the previously published Bouguer gravity anomaly map of Cyprus[1], reproduced here as Fig. 2. This strong correlation between the gravity and magnetic fields, and the systematic association of pillow lava outcrops with positive magnetic anomalies, suggest that at least the pillow lava units of the igneous massif are normally magnetized throughout and that the magnetic anomalies can be explained in terms of the outcrop pattern of the Massif and the form of its buried extent; the latter being revealed by the

Table 1 Mean Magnetic Properties for the Various Units of the Troodos Igneous Massif (see also ref. 4)

(a) Mean directions of natural remanent magnetization (NRM) derived from a Fisher analysis of site means

Rock unit	Dec.	Incl.	N	R	k	α_{95}
Upper Pillow lavas	254.9	38.6	3	2.562	4.6	66.0
Lower Pillow lavas	290.5	35.1	4	3.727	11.0	29.0
All Pillow lavas	276.5	37.8	7	6.103	6.7	25.2
Greenstone dykes	316.1	55.4	5	4.686	12.7	22.3
Gabbros and diorites	267.7	61.9	10	8.403	5.6	22.3
Pyroxenite and dunite	328.6	68.3	6	5.187	6.2	29.3
Harzburgite and serpentinite	354.9	42.8	3	2.531	4.3	69.1
			Total 31			

(b) Mean values of NRM intensity, volume susceptibility and Koenigsberger ratio (Q_n) derived from sample means

Rock unit	$NS(n)$	NRM intensity Mean $\times 10^3$	NRM intensity Log mean	NRM intensity Log s.d.	Susceptibility Mean $\times 10^3$	Susceptibility Log mean	Susceptibility Log s.d.	Q_n Mean	Q_n Log mean	Q_n Log s.d.
UPL	18 (36)	1.16	−2.94	0.36	0.56	−3.25	0.74	4.68	0.67	0.49
LPL	36 (75)	7.18	−2.14	0.50	2.78	−2.56	0.20	5.79	0.76	0.46
All PL	54 (111)	3.91	−2.41	0.59	1.63	−2.79	0.56	5.40	0.73	4.37
Dykes	41 (115)	0.76	−3.12	0.67	1.69	−2.77	0.63	1.01	0.00	0.23
Gabbros/dior.	55 (127)	0.55	−3.26	0.59	0.25	−3.61	0.73	4.99	0.70	0.54
Pyrox./dun.	29 (73)	6.47	−2.19	0.39	1.55	−2.81	0.47	9.30	0.97	0.42
Harz./serp.	18 (49)	0.20	−3.70	0.63	0.32	−3.50	0.40	1.42	0.15	0.53
Total	197 (475)									

Dec., declination east of north; Inc., inclination with respect to horizontal, positive downwards; N, number of sites; R, resultant of N unit vectors; k, precision parameter; α_{95} = semi-angle of 95% confidence cone; NS, number of samples; (n) = number of specimens; NRM, intensity in e.m.u. or gauss cm^{-3}; susceptibility is volume susceptibility cm^{-3}; Q_n, ratio of NRM intensity to induced magnetization (=0.45 × susceptibility, where 0.45 = ambient magnetic field intensity in gauss); mean, mean of log normal distributions (after ref. 22); all logs to the base 10; s.d., standard deviation.

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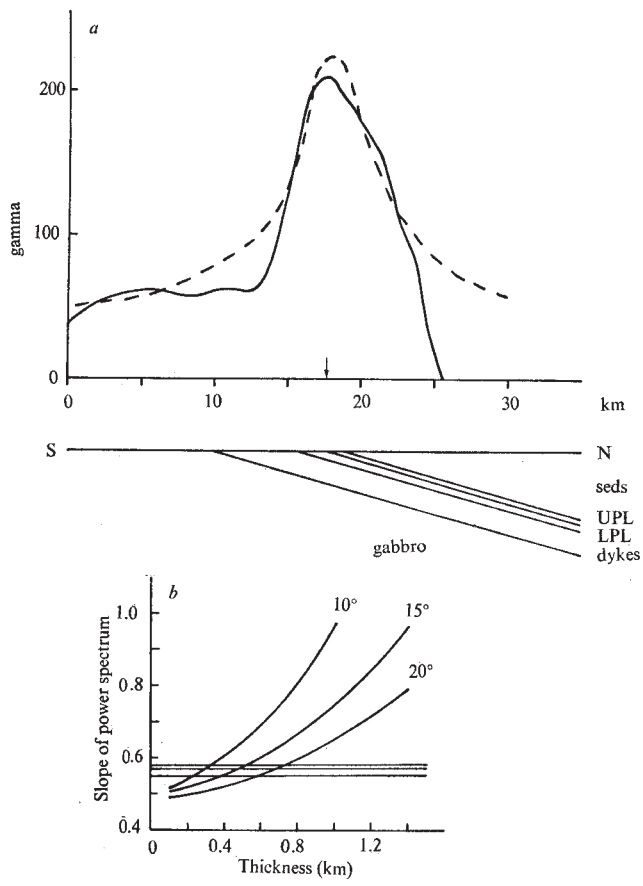


Fig. 3 *a*, An observed magnetic anomaly profile across the northern limb of the Massif (——) compared with a computed anomaly profile (- - -) assuming the model shown and magnetic properties for the various units as given in Table 1. Azimuth of profiles = 020°. Declination, dip and intensity of Earth's magnetic field taken as 2° E, +50°, and 45,000 gamma respectively. The profiles were matched at the Upper/Lower Pillow Lava contact (indicated by arrow). *b*, Plot of the thickness assumed for the Lower Pillow Lava unit, in simulations comparable to that shown above, against the negative slope of a straight line fitted to the log of the power in the lower five harmonics of the computed profile's power spectrum[16]. The dip of the unit was varied through 10°, 15° and 20°. The equivalent power spectra slopes for three observed profiles, which were corrected by upward continuation for any source-to-sensor distance variations, are indicated by the horizontal lines.

gravity field. This proposal is consistent with the directions of remanent magnetization measured in orientated hand samples and drill cores removed from the Troodos outcrop area and measured in the field and in the laboratory respectively[3]. These measurements, on specimens from 150 localities well distributed within the complex, yielded no evidence for reversed remanent magnetization in any of the igneous rock units. Further, in eastern Cyprus, over the buried portion of the Massif, the turning points and gradients on the magnetic anomaly map are systematically offset to the east with respect to those on the Bouguer anomaly map. This is consistent with the direction of natural remanent magnetization measured in the pillow lavas of the Troodos complex which has a large westerly component of declination, and a dip of approximately 35° down (Table 1). It seems probable therefore that the magnetic anomalies observed over both the exposed and buried portions of the Troodos Massif can be explained without invoking reversed magnetization.

As noted previously, the most pronounced magnetic anomalies are highs associated with the Troodos Pillow Lava Series, either where it is exposed around the perimeter of the Massif, or where it is known to be at or near the surface, as in the vicinity of the Troulli inlier, to the east of the main outcrop area, and beneath Larnaca and Famagusta on the east-facing coasts. Recent metamorphic data[12] convincingly support the proposal[2] that the lower division of the Pillow Lava Series was formed at a spreading axis whereas the Upper Pillow Lavas are an off axis sequence analogous perhaps to the seamounts of the present day ocean floor. Further, the most important contribution to the magnetic anomalies comes from the strongly magnetized Lower Pillow Lavas (Table 1). Because several of the most prominent positive magnetic anomalies overlie known massive sulphide deposits, it is tempting to explain this coincidence in terms of a common parameter, especially as the sulphides are of possibly sedimentary origin and lie in depressions in the Lower Pillow Lava surface[13]—a geological environment analogous to that of the "hot brine holes" in the Red Sea median trough[14]. As tensional faulting parallel to the ridge axis during seafloor spreading processes produces rugged topography, the bathymetric highs would be subjected to more intense submarine erosion and the uppermost extrusive layer would become thinner than in adjacent basins and troughs. It seems therefore that the most ready explanation of the partial coincidence of positive magnetic anomalies with massive sulphide deposits is that the thickest Lower Pillow Lava sequences, producing the magnetic highs, underlie the bathymetric depressions. This tentative conjecture must await more precise information concerning the space form of the sulphide masses and the thicknesses of the underlying pillow lava sequences.

Other Anomalies

In addition to those associated with the Pillow Lava Series, a pronounced anomaly occurs above Mount Olympus, which is roughly central within the principal Troodos outcrop area. This anomaly seems to be associated with the ultramafic rocks of the summit area. To the east these consist of highly serpentinized and tectonized peridotites which presumably form part of the serpentinite diapir thought to underlie Mount Olympus because of the intense and localized negative gravity anomaly centred on it[1]. Similar highly serpentinized ultramafics outcrop in the Limassol Forest to the south-east; both areas are characterized by a negative magnetic anomaly. The harzburgites and serpentinites on the eastern flanks of Mount Olympus are considered to represent depleted and remobilized mantle respectively[3,5]. In contrast, the less serpentinized pyroxenites, peridotites and dunites of the western slopes have been interpreted as cumulate phases precipitated from a basaltic melt produced by the partial fusion of the Earth's mantle[3,5]. The ultramafic rocks of cumulate origin have very high intensities of magnetization, comparable to those of the Lower Pillow Lavas and typically an order of magnitude higher than those of the other units of the Massif (see Table 1). It is not surprising therefore that the outcrop pattern of these units is marked by a pronounced positive magnetic anomaly whereas the outcrop area of the remaining igneous units—the altered basic dykes of the Sheeted Intrusive Complex and the gabbros and more silicic rocks of the Plutonic Complex—is essentially featureless.

Along the northern margin of the main outcrop area of the Troodos Massif the units of the Complex dip gently at 10–15° beneath the sediments of the Mesaoria plain. On the basis of this dip and the mean magnetic properties of the units of the Massif given in Table 1, computed and observed anomaly wave forms along north–south lines have been compared using spectral analysis. The chief variable in this analysis is the thickness of the Lower Pillow Lavas although variations in dip were also considered. The analysis confirms that the thickness of the strongly magnetized pillow lavas is almost certainly no greater than 0.5 km as suggested previously on the basis of work on Cyprus[4] and at sea[15]. An example of the simulation of one of these profiles, assuming that the Lower Pillow Lavas have a thickness of 0.5 km, is shown in Fig. 3.

In the eastern part of Cyprus, to the east of the igneous outcrop, there is a pronounced negative magnetic anomaly of

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maximum amplitude 500 gamma and north–south trend (Fig. 1). The form, amplitude and wavelength of this anomaly are comparable to those of the distinctive linear anomalies observed at sea. Moreover, the trend of the anomaly is parallel to the predominant north–south strike direction of the dyke complex (Sheeted Intrusive Complex) of the Massif and at right angles to a pronounced linear structure running due east–west, to the north of the Limassol Forest, which has been interpreted[3] as a fossil transform fault. The anomaly is therefore parallel to the structural grain of the postulated ocean floor and at right angles to the inferred spreading direction, as are linear magnetic anomalies at sea. It occurs over an area where, from borehole evidence, pillow lavas of the Troodos Massif are known to occur at depth beneath mainly Tertiary sediments. On the basis of this information, it is tempting to assume that the anomaly arises from the occurrence of a north–south strip of reversely magnetized material within the unexposed portion of the Troodos Igneous Complex.

But this linear anomaly correlates very strongly with a similar disturbance in the gravity field and it seems probable that both can be explained in terms of a single feature within the buried portion of the Troodos complex. If it is assumed that the gravity anomaly arises from a graben-like depression in the topography of the igneous basement, the form of this depression may be deduced from the gravity anomaly if appropriate density contrasts between the igneous rock units and overlying sediments are assumed. A possible configuration for the basement, deduced in this way, is illustrated in Fig. 4. If the magnetic properties given in Table 1 are then assumed for the various units of the complex the magnetic anomaly arising from this basement topography can also be calculated (Fig. 4). The computed magnetic anomaly reproduces the characteristic eastward displacement of turning points and gradients with respect to those of the gravity anomaly curve but its amplitude is little more than one-third of the amplitude of the observed magnetic anomaly. Closer agreement between the observed and simulated magnetic anomalies could be obtained by assuming higher intensities of magnetization for the igneous units or lower density contrasts between the igneous and sedimentary rocks, which would imply a deeper depression. But the magnitude of the changes in either or both of these sets of parameters, which would be necessary to eliminate this discrepancy, renders them very unlikely. It could be argued that, having satisfied the gravity anomaly in this way, the magnetic anomaly may still arise from normal/reverse magnetization contrasts within the basement complex and resulting from its formation at a mid-ocean ridge crest. If this is the case it can be argued that the essentially vertical contrasts between normally and reversely magnetized material will only be present in the Lower Pillow Lava layer and the dyke complex[4]. The Upper Pillow Lava series is thought to have been emplaced off-axis at a later date and the plutonic units are thought to have acquired their magnetization in such a way as to produce sub-horizontal contrasts which are rather ineffectual as regards the production of magnetic anomalies. In any event the most important contribution to the magnetic anomaly will come from the strongly magnetized Lower Pillow Lava layer which is certainly thought to have formed at a ridge crest. The only other strongly magnetized unit is the layer of ultramafic accumulates which presumably acquired its present magnetic properties as a result of serpentinization after its emplacement at the ridge crest; it will not therefore preserve a record of normal/reverse magnetization, related to spreading and reversals of the Earth's magnetic field. A magnetic anomaly simulation which assumes vertical normal/reverse magnetization contrasts within the Lower Pillow Lava and dyke complex units is illustrated in Fig. 4. In contrast to the earlier simulation the computed anomaly amplitude is now somewhat too high and the computed anomaly gradients too steep.

An alternative explanation of these essentially coincident magnetic and gravity anomalies is that they result from a zone of less dense and less magnetic material within the igneous complex. Such a zone may arise from the fracturing and brecciation of the complex or the intrusion of serpentinite within it. In this way one can produce intrabasement density and magnetization contrasts capable of producing the observed gravity and magnetic anomalies without recourse to large scale topographic relief on the igneous basement or to reversed magnetization. Preliminary simulations of the gravity and magnetic anomalies based on a simple model of this type are illustrated in Fig. 5. The simulation of the magnetic anomaly profile is

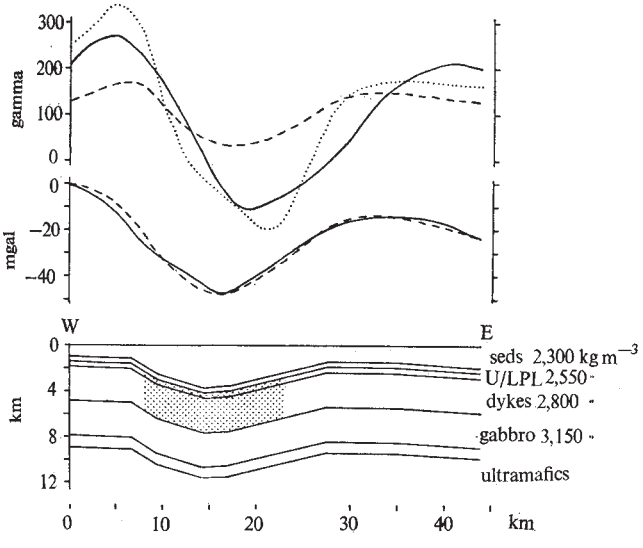


Fig. 4 Observed magnetic and gravity anomaly profiles across the pronounced negative anomaly in both fields over eastern Cyprus (see Figs 1 and 2) are indicated by solid curves and compared with calculated profiles (dashed curves) derived from the model shown. The magnetic properties assumed for the various igneous rock units are given in Table 1, and the densities assumed are given on the diagram. The calculated magnetic anomaly profile indicated by dots results from assuming that the shaded volume of the Lower Pillow Lavas and Sheeted (dyke) Complex indicated on the model has reversed natural remanent magnetization (see text).

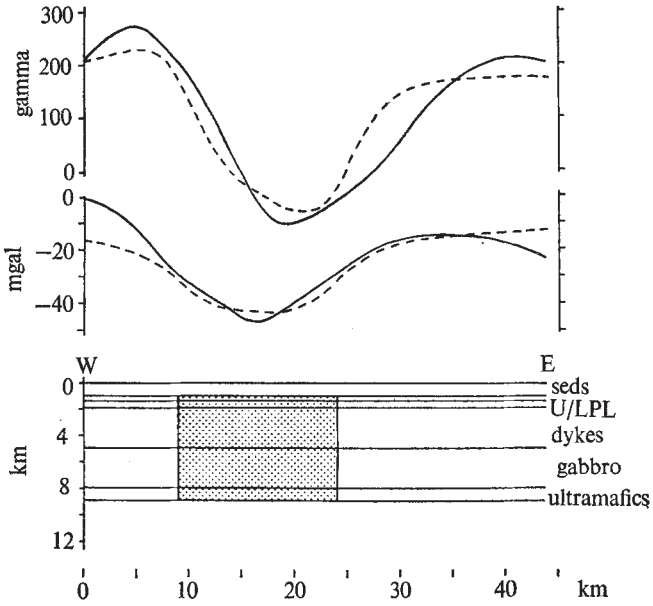


Fig. 5 The observed magnetic and gravity anomaly profiles of Fig. 4 compared with calculated profiles derived from an alternative model. This model assumes no relief on the igneous rock units but a zone, indicated by shading, in which the density and magnetization of these units are reduced compared to that assumed in Fig. 4. At the vertical boundaries of this zone the magnetization contrast is 1 : 3 and the density contrast 150 kg m^{-3}.

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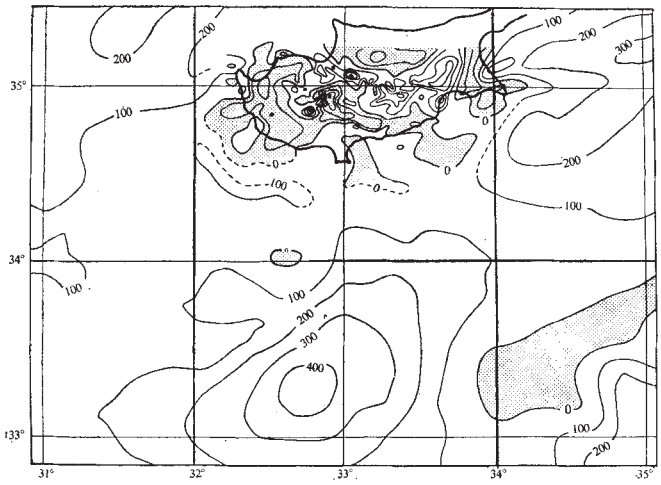


Fig. 6 Total field aeromagnetic anomaly map for part of the eastern Mediterranean. The map was derived from the survey presented here (Fig. 1) and the marine survey published by Woodside and Bowin[20]. An appropriate regional field obtained from the IGRF (ref. 11) was subtracted from the latter and the resulting anomaly map continued upwards to the level of the aeromagnetic survey of Cyprus so that the two maps could be integrated. Contour interval=100 gamma. Areas of negative anomaly are shaded. Contour lines are dashed where control is poor or where values are indefinite because of edge effects in the upward continuation process.

the best of the three considered here (Figs 4 and 5). Clearly the interpretation of these anomalies is ambiguous. But it seems unlikely that the association of anomalies in both potential fields is entirely fortuitous and one hesitates therefore to invoke reversed remanent magnetization in the igneous basement despite the appropriate form and orientation of the magnetic anomaly.

Relation to Magnetic Reversals

Neither this survey nor magnetic measurements on the outcropping rocks have produced convincing evidence for reversely magnetized material within the Troodos Massif. This is surprising, for if the Massif was formed by seafloor spreading at a rate of 1–2 cm yr^{-1} (ref. 3), then the across-strike distance covered by these surveys represents 10–15 m.y. of spreading history. Previously the age of the Troodos Massif has been assumed to be pre-Campanian on the basis of radiolarian assemblages in the sediments immediately overlying the pillow lavas[3]. In Table 2 we present, for the first time (thanks to D. C. Rex and M. H. Dodson of Leeds University), some potassium-argon age determinations on what were considered to be the most suitable materials, for dating purposes, from the Massif. These determinations confirm the pre-Campanian age (76+ m.y.) although two seem to be anomalously low.

It is perhaps significant that syntheses of palaeomagnetic studies of rocks of Cretaceous age suggest that there was a period, of at least 10 m.y. duration, immediately before 75–80 m.y. BP during which the Earth's magnetic field was of a single, normal polarity[17–19]. Fortuitous and perverse as it may seem, it could well be that the Troodos Massif was formed by spreading at that time and that it does not therefore preserve within it avenues of normally and reversely magnetized material. The absence of these avenues in the Troodos Massif does not therefore preclude the possibility that it was formed by seafloor spreading but is disappointing because had they been present it would have proved, virtually unequivocally, that the Massif was formed by this process. Also, the detailed interfingering relationships of normally and reversely magnetized material would have yielded invaluable structural information on the process of the formation of oceanic crust at ridge crests. Thus, although the petrology, structure and age of the Troodos Massif are entirely consistent with the suggestion that it is a fragment of oceanic lithosphere formed by the process of seafloor spreading during mid- to late Cretaceous the disturbances in the Earth's magnetic field over Cyprus can be explained in terms of the outcrop patterns of the Troodos igneous Massif, and the form of its buried extent, without necessarily invoking reversed magnetization of any part of it. This is particularly relevant when the magnetic field in the eastern Mediterranean area as a whole is considered.

Table 2 K-Ar Age Determinations

Rock type	Wt % K	Vol ^{40}Ar rad. $cm^3\ g^{-1} \times 10^6$	% ^{40}Ar rad.	Age m.y.
UPL limburgite (whole rock)	0.52	1.863	70.5	88±4
		1.880	68.5	89±4
UPL trachyandesite (whole rock)	1.33	3.159	61.8	59±3
		3.027	60.6	56±3
Quartz diorite (whole rock)	0.37	1.138	58.7	76±3
		1.201	52.3	79±3
Pegmatitic gabbro (hornblende)	0.094	0.251	10.2	66±7
		0.258	18.8	67±7

Decay constants for ^{40}K: $\lambda_e = 0.584 \times 10^{-10}\ yr^{-1}$; $\lambda\beta = 4.72 \times 10^{-10}\ yr^{-1}$.
Atomic abundance of $^{40}K/K_{total} = 1.19 \times 10^{-4}$ atoms/atom.

The area covered by the aeromagnetic survey presented here and shown in Fig. 1 was chosen so that it would link up with surveys of the Earth's magnetic field over adjoining areas. In Fig. 6 the anomaly map of Fig. 1 has been integrated with the marine magnetic survey compiled by Woodside and Bowin[20] to produce an anomaly map covering most of the eastern Mediterranean. As noted previously[21], the eastern Mediterranean is remarkably devoid of oceanic-type linear magnetic anomalies and it is therefore debatable whether it is underlain by oceanic crust formed by seafloor spreading. In view of the Cyprus evidence, however, it is tempting to suggest that some of the crust beneath the eastern Mediterranean might well be oceanic, and formed by the process of seafloor spreading during the mid to late Cretaceous when the Earth's magnetic field was of constant and normal polarity.

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